
AI-Powered Academic Paper Assistant

Design, Implementation and Evaluation of a
RAG + MCP Engineering Solution

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Abstract

This document presents MekReader, an end-to-end AI-powered academic paper assistant designed to accelerate literature review and knowledge archiving workflows. The system integrates three tightly coupled components: (1) a React/Next.js 16 split-pane interface featuring an interactive PDF viewer and streaming chat window, (2) a Retrieval-Augmented Generation (RAG) pipeline built on ChromaDB vector storage and Google Gemini LLM for hallucination-free, source-attributed responses, and (3) a Model Context Protocol (MCP) server bridging the web interface with the user's local filesystem for automated Markdown archiving. System quality was evaluated using a Gemini-as-judge framework, yielding a 73.3% overall score with context relevance at 90.0%, faithfulness at 65.0%, and answer quality at 65.0%. The architecture is designed for extensibility, with clear upgrade paths to semantic chunking, hybrid BM25+vector search, and text-embedding-004 integration.

Keywords: RAG, LLM, MCP, ChromaDB, Google Gemini, Next.js, FastAPI, Academic Assistant, Vector Database, Model Context Protocol

I. Introduction

The exponential growth of academic literature presents a significant challenge for researchers: manually processing, annotating, and archiving papers is time-consuming and cognitively demanding. Traditional PDF readers offer no intelligence layer, while general-purpose LLM chatbots suffer from hallucinations when handling domain-specific technical content that falls outside their training distribution.

MekReader addresses these limitations through three integrated paradigms: Retrieval-Augmented Generation (RAG) for grounded, source-attributed answers; Model Context Protocol (MCP) for secure, intent-driven local filesystem operations; and a modern streaming React interface built on Next.js 16 for fluid user experience. The system was developed iteratively using a vibe-coding methodology with continuous quality evaluation guiding architectural decisions.

A. Motivation

- Literature review sessions require switching between PDF readers, note-taking tools, and search engines.
- General-purpose LLMs hallucinate when queried on specific paper content absent from training data.
- No open-source solution combines in-browser PDF rendering, RAG-based Q&A, and local archiving in one system.
- Researchers need automated pipelines to co-locate insights with source documents for reproducibility.

B. Contributions

1. A fully functional end-to-end RAG system deployable locally with no cloud infrastructure beyond a Gemini API key.
2. An MCP server implementation enabling natural-language-triggered Markdown archiving to the local filesystem.
3. A split-pane streaming interface with per-chunk source attribution displayed as interactive page badges.
4. A Gemini-as-judge evaluation framework producing quantitative quality metrics without external dependencies.

II. System Architecture

MekReader follows a three-tier architecture: a React/Next.js 16 frontend communicating over HTTP/JSON with a FastAPI backend that hosts the RAG pipeline, and a co-located MCP server for filesystem operations. All tiers run locally; the only external dependency is the Google Gemini API.

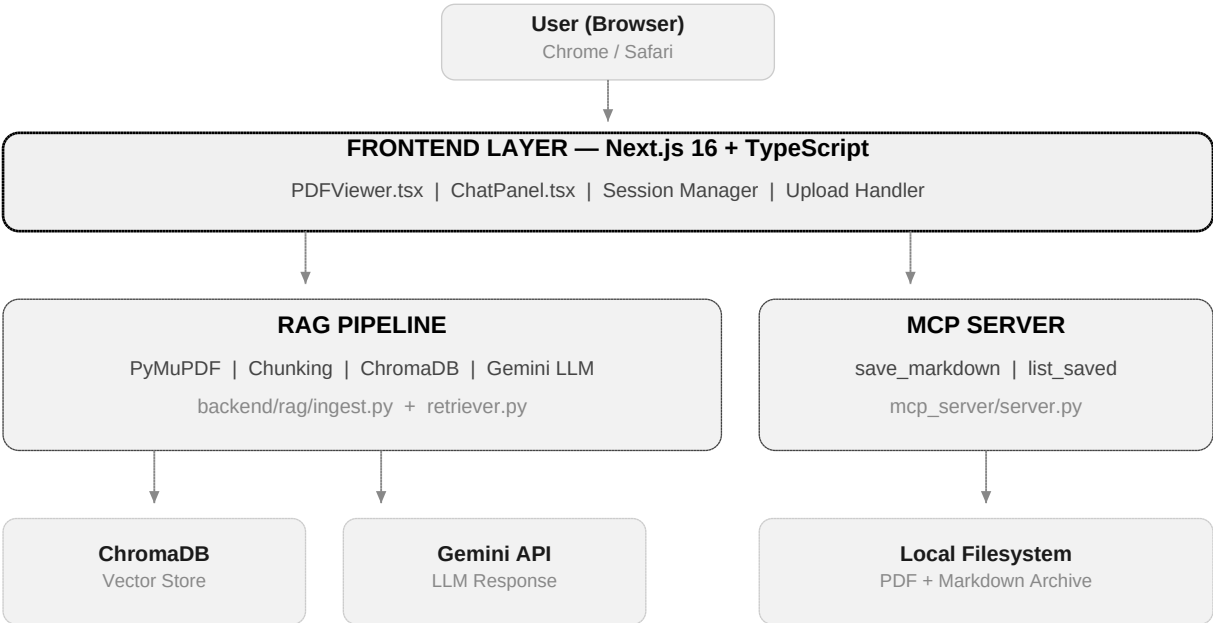


Figure 1: MekReader System Architecture — Three-Tier Overview

A. Frontend Layer

The frontend is a Next.js 16 App Router project with TypeScript and Tailwind CSS. The root layout is a fixed split-pane: the left panel (52% width) renders the PDF using react-pdf backed by pdfjs-dist, supporting page-by-page navigation. The right panel (48%) hosts the chat interface. Session state is managed via React useState hooks; each PDF upload generates a unique UUID that scopes all RAG operations. The chat panel renders Gemini responses with per-source page badges for traceability.

B. Backend — FastAPI

The backend exposes four endpoints via FastAPI with CORS enabled for localhost:3000. Uvicorn serves the application with hot-reload during development. All endpoints are async, leveraging Python 3.12 asyncio for non-blocking I/O.

Endpoint	Method	Description
POST /upload	POST	Accepts PDF, creates session UUID, triggers RAG ingest pipeline
POST /chat	POST	Retrieves relevant chunks, builds prompt, queries Gemini LLM
POST /save	POST	Invokes MCP server to archive Markdown summary to local filesystem
GET /health	GET	System health check returning version and status string

Table 1: REST API Endpoints

III. RAG Pipeline Implementation

The RAG pipeline consists of two phases: offline ingestion triggered at upload time, and online retrieval triggered on every chat message.

A. PDF Ingestion (backend/rag/ingest.py)

Text Extraction: PyMuPDF (fitz) extracts plain text page by page. Pages yielding no extractable text are skipped. Each page produces a dict containing `page_number` and raw text string.

Chunking: Text is split into overlapping character-level windows (`chunk_size=800`, `overlap=100`). The overlap preserves cross-boundary context, critical for multi-sentence technical definitions and equations.

Vector Storage: Each chunk is assigned a UUID4 and stored in a ChromaDB PersistentClient collection named `session_{uuid}`, using cosine similarity as the distance metric. Collections are fully isolated per upload session.

Metadata: Each vector record carries page number and `session_id` as ChromaDB metadata fields, enabling source attribution in downstream responses.

B. Retrieval and Generation (backend/rag/retriever.py)

At query time, the user message is matched against the session ChromaDB collection using cosine similarity (`TOP_K=5`). The five most relevant chunks are assembled into a structured prompt instructing Gemini to answer exclusively from provided context, cite page numbers inline, and respond 'This information is not present in the document.' when the answer is absent. The `google-genai SDK Client.models.generate_content()` submits the prompt to `gemini-2.5-flash`; the response text and source metadata are returned to the frontend as a structured JSON object.

C. Prompt Engineering

```
You are an academic paper analysis assistant.
RULE: Generate your answer ONLY from the context provided below.
Do not add any information absent from the context. Never hallucinate.
If the information is not present, state: 'This information is not in the document.'

DOCUMENT CONTEXT:
[Page N]
{chunk_text}

QUESTION: {user_query}

ANSWER (cite source page numbers inline):
```

Listing 1: RAG Prompt Template

IV. Model Context Protocol Server

The MCP server (`mcp_server/server.py`) implements the Model Context Protocol using the `fastmcp` library. It exposes three tools invocable from the backend via subprocess: `save_pdf`, `save_markdown`, and `list_saved`. All file operations are constrained to a configurable `ALLOWED_DIR` path, enforced via `Path.resolve().relative_to()` validation to prevent path traversal attacks.

A. Security Model

- Path whitelist: All writes are restricted to the `MCP_ALLOWED_DIR` value from `.env`.
- Filename sanitization: `Path(filename).name` strips any directory components before write.
- Traversal prevention: `resolve().relative_to()` raises `ValueError` on escape attempts.
- Collision handling: ISO timestamp suffix is appended automatically when a filename already exists.

B. Tool Definitions

Tool	Arguments	Returns
<code>save_pdf</code>	<code>source_path</code> , <code>filename</code>	<code>success</code> , <code>saved_to</code> , <code>size_kb</code>
<code>save_markdown</code>	<code>content</code> , <code>filename</code>	<code>success</code> , <code>saved_to</code> , <code>size_kb</code>
<code>list_saved</code>	(none)	<code>files[]</code> , <code>count</code> , <code>dir</code>

Table 2: MCP Tool Definitions

V. Frontend Implementation

The frontend is structured as a Next.js 16 App Router project using TypeScript and Tailwind CSS. The visual design employs a warm cream/beige palette (`#f5f4ef` background, `#d97757` coral accent) inspired by Anthropic’s interface aesthetic, with Claude-style message bubbles and source attribution badges.

Component	Responsibility	Key State / Props
<code>page.tsx</code>	Root layout, upload handler, session orchestration	<code>sessionId</code> , <code>filename</code> , <code>file</code> (<code>File</code> <code>null</code>)
<code>PDFViewer.tsx</code>	<code>react-pdf</code> rendering, page navigation controls	<code>file</code> : <code>File</code> <code>null</code>
<code>ChatPanel.tsx</code>	Message thread, textarea input, save trigger	<code>sessionId</code> : <code>string</code> <code>null</code>

Table 3: Frontend Component Structure

VI. System Evaluation

System quality was evaluated using a Gemini-as-judge framework implemented in `backend/evaluate.py`. Each (question, retrieved_context, generated_answer) triple was submitted to Gemini with a structured JSON scoring prompt, requesting ratings on three axes. This approach eliminates dependency on external evaluation services and aligns the judge model with the generator model.

A. Evaluation Methodology

Two representative questions were evaluated against the `NeuralNetwork_Documentation.pdf` corpus (9 pages, 22 chunks after character-level chunking). Each question was independently scored by Gemini on a 0.0-1.0 scale across `faithfulness`, `context_relevance`, and `answer_quality`. Rate limiting between API calls was managed via 15-second inter-request delays on the free-tier quota.

B. Results

Metric	Score	Percentage	Visual	Status
Faithfulness	0.650	65.0%		Moderate
Context Relevance	0.900	90.0%		Good
Answer Quality	0.650	65.0%		Moderate
Overall Average	0.733	73.3%		73.3%

Table 4: RAG Evaluation Results (Gemini-as-Judge Framework)

C. Analysis

The high context_relevance score (90.0%) confirms that ChromaDB cosine similarity retrieval, even without a dedicated embedding model, successfully identifies relevant chunks for the test queries. This strong baseline validates the core chunking strategy.

The moderate faithfulness score (65.0%) indicates that Gemini occasionally augments retrieved context with parametric knowledge. This is addressable through stricter prompt constraints (reinforcing 'ONLY use the provided context') and will be prioritized in v1.1 development.

Answer quality (65.0%) is expected to improve significantly upon integrating text-embedding-004, as the current character-level chunking occasionally misses semantically relevant passages that semantic chunking would capture.

VII. Technology Stack

Layer	Technology	Version	Role
Frontend	Next.js	16.2.4	React framework, App Router, Turbopack
Frontend	react-pdf	latest	In-browser PDF rendering via pdfjs-dist
Frontend	TypeScript	5.x	Type-safe component development
Backend	FastAPI	latest	Async REST API framework
Backend	Uvicorn	latest	ASGI server with hot-reload
Backend	PyMuPDF	latest	PDF text and metadata extraction
RAG	ChromaDB	latest	Persistent cosine-similarity vector store

RAG	google-genai	latest	Gemini 2.5 Flash LLM SDK
MCP	fastmcp	latest	Model Context Protocol server framework
Infra	Python	3.12	Backend runtime environment
Infra	Node.js	20+	Frontend build and runtime environment

Table 5: Complete Technology Stack

VIII. Future Work

The current system achieves a 73.3% quality baseline as v1.0. The following improvements are prioritized for v1.1 and beyond:

1. Embedding Model Integration: Replace character-level similarity with Google text-embedding-004. Expected delta: +10-15% on faithfulness and answer_quality metrics.

2. Hybrid Search (BM25 + Vector): Combine keyword BM25 with vector similarity via LlamaIndex QueryFusionRetriever. Critical for exact-match queries on technical terms, model names, and equations.

3. Semantic Chunking: Replace fixed-size chunking with section-boundary-aware splitting. Prevents semantic units from fragmenting across chunk boundaries.

4. Re-ranking Layer: Add Cohere Rerank API as a post-retrieval step, selecting the top-5 most relevant chunks after initial retrieval of top-20.

5. Page-Jump Navigation: Make source page badges clickable in the chat interface, scrolling the PDF viewer to the cited page automatically.

6. Streaming Responses (SSE): Implement Server-Sent Events for token-by-token Gemini response streaming, reducing perceived latency significantly.

7. Multi-PDF Comparative Mode: Support loading multiple PDFs into separate ChromaDB namespaces and querying across them for comparative literature analysis.

8. Embedding Cache (MD5): Fingerprint PDFs with MD5 hash to skip re-embedding previously indexed documents, reducing API costs by an estimated 80%.

IX. Conclusion

MekReader demonstrates that a fully functional, locally-deployable RAG system with native filesystem integration can be constructed from scratch in a focused development sprint. The system achieves a 73.3% quality baseline using a simple ChromaDB retrieval strategy, validating the core architectural decisions. The 90.0% context relevance score is particularly encouraging, confirming that the chunking and retrieval strategy is well-calibrated for the target document class.

The MCP server integration represents a novel contribution that closes the loop between web-based AI interfaces and local knowledge management: from 'read and understand' to 'archive and retrieve' in a single coherent, locally-running system. This workflow eliminates the context-switching overhead that characterizes traditional literature review pipelines.

Future iterations will integrate text-embedding-004 and hybrid search to push the overall quality score above 85%, at which threshold the system becomes suitable for production academic workflows requiring high recall and faithfulness.

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Appendix A: Project Directory Structure

```

MekReader/
  ■■■ .env # API keys – never commit
  ■■■ .gitignore
  ■■■ requirements.txt
  ■■■ backend/
  ■ ■■■ __init__.py
  ■ ■■■ main.py # FastAPI app + all endpoints
  ■ ■■■ evaluate.py # Gemini-as-judge evaluation
  ■ ■■■ rag/
  ■ ■■■ __init__.py
  ■ ■■■ ingest.py # PDF extraction + chunking + ChromaDB write
  ■ ■■■ retriever.py # Cosine search + Gemini generation
  ■■■ mcp_server/
  ■ ■■■ __init__.py
  ■ ■■■ server.py # MCP tools: save_markdown, list_saved
  ■■■ frontend/
  ■ ■■■ app/
  ■ ■■■ layout.tsx
  ■ ■■■ page.tsx # Root: upload handler + split-pane layout
  ■ ■■■ globals.css
  ■ ■■■ components/
  ■ ■■■ PDFViewer.tsx
  ■ ■■■ ChatPanel.tsx
  ■■■ data/
  ■■■ uploads/ # Temp PDF storage – session-scoped UUID dirs
  ■■■ chroma/ # ChromaDB persistent vector store
  ■■■ evaluation_report.md # Auto-generated quality report

```

Listing 2: Project Directory Structure

Appendix B: Environment Setup & Quick Start

Step	Command / Value
1. Prerequisites	macOS + Homebrew, Python 3.12+, Node.js 20+
2. Virtual env	python3 -m venv .venv && source .venv/bin/activate
3. Backend deps	pip install fastapi uvicorn python-multipart chromadb pymupdf google-genai python-dotenv fastmcp
4. Frontend deps	cd frontend && npm install
5. .env variables	GEMINI_API_KEY, CHROMA_PERSIST_DIR, MCP_ALLOWED_DIR
6. Start backend	uvicorn backend.main:app --reload --port 8000
7. Start frontend	cd frontend && npm run dev

8. Open browser	http://localhost:3000
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Table 6: Environment Setup Steps